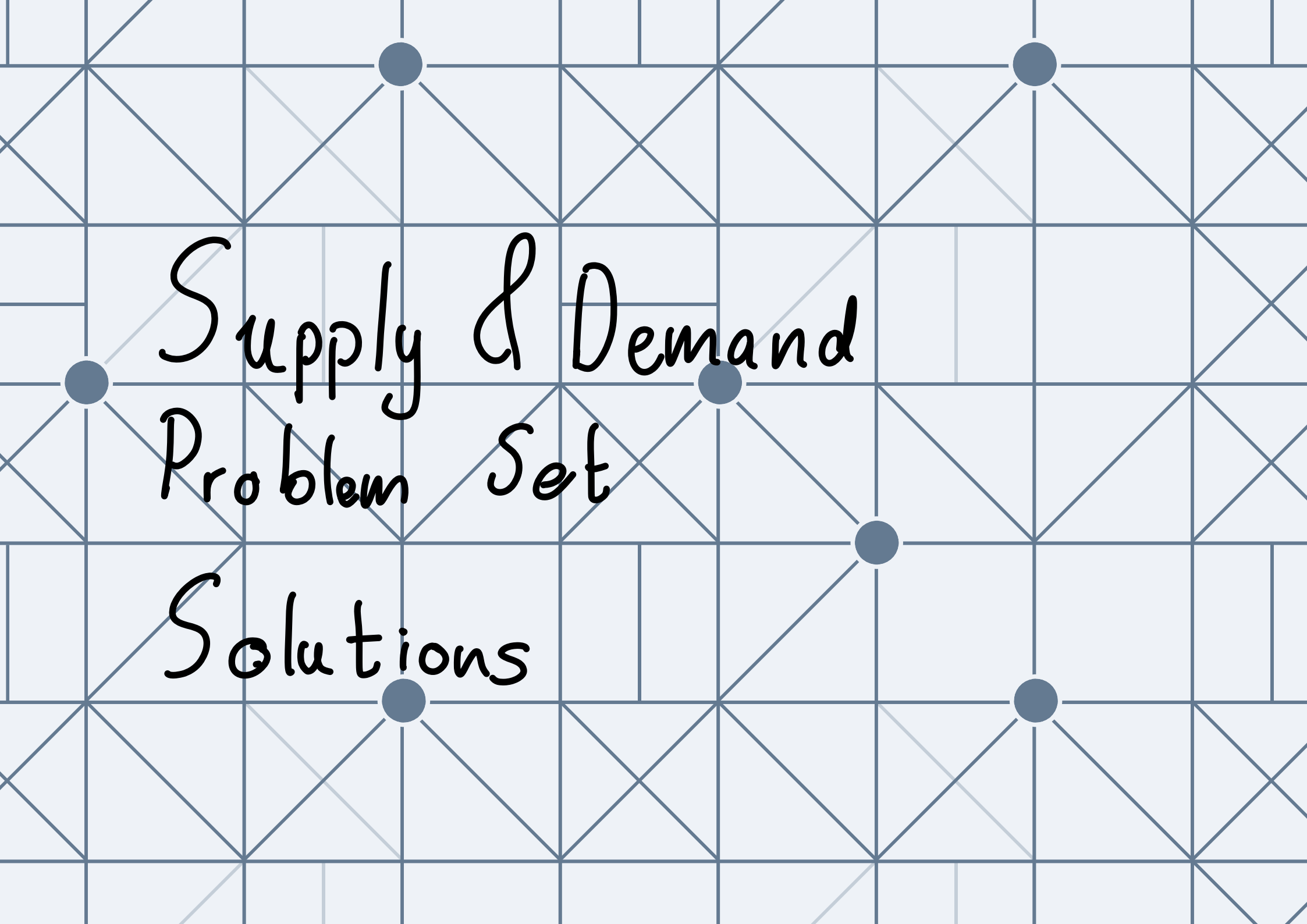




Supply & Demand
Problem Set

Solutions

Problem 1

$$Q_{choc}^D = 560 - 4P_{choc} + 0.1Y + 4P_{caramel} - 5P_{milk}$$

$$Q_{choc}^S = -60 + 6P_{choc} - 10P_{cacao}$$

The value of all determinants except Y is given.

Y is determined in the labor market, i.e., $Y = w^*L^*$ where (w^*, L^*) is the labor market equilibrium wage and quantity of labor.

Eq. in the labor market: $100 - 2w^* = -20 + 4w^*$
 $\Rightarrow 120 = 6w^* \Rightarrow w^* = 20$
 $L^* = 100 - 2(20) = 60$
 $Y = w^*L^* = 1200$

Eq. in the chocolate market:

$$Q_{choc}^D = 560 - 4P_{choc} + 0.1(1200) + 4(50) - 5(8)$$

$$= 560 - 4P_{choc} + 120 + 200 - 40$$

$$= 840 - 4P_{choc}$$

$$Q_{choc}^S = -60 + 6P_{choc} - 10(10) = -160 + 6P_{choc}$$

For the mkt. equilibrium (Q_{choc}^*, P_{choc}^*) we must have:

$$Q_{choc}^D(P_{choc}^*) = Q_{choc}^S(P_{choc}^*)$$

$$840 - 4P_{choc}^* = -160 + 6P_{choc}^*$$

$$\Rightarrow 1000 = 10P_{choc}^*$$

$$\Rightarrow P_{choc}^* = 100 \quad Q_{choc}^* = 840 - 400 = 440 //$$

Problem 2

$$1. Q_{d1} = 200 - P_1$$

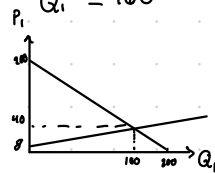
$$Q_{s1} = -40 + 5P_1$$

$$200 - P_1^* = -40 + 5P_1^*$$

$$\Rightarrow 240 = 6P_1^*$$

$$\Rightarrow P_1^* = 40$$

$$Q_1^* = 160$$



$$Q_{d2} = 100 - 2P_2$$

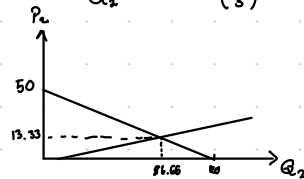
$$Q_{s2} = 20 + 4P_2$$

$$100 - 2P_2^* = 20 + 4P_2^*$$

$$80 = 6P_2^*$$

$$\Rightarrow P_2^* = \frac{80}{3} \approx 13.33$$

$$Q_2^* = 100 - 2\left(\frac{80}{3}\right) = \frac{300 - 160}{3} = \frac{140}{3} \approx 46.66$$



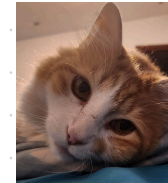
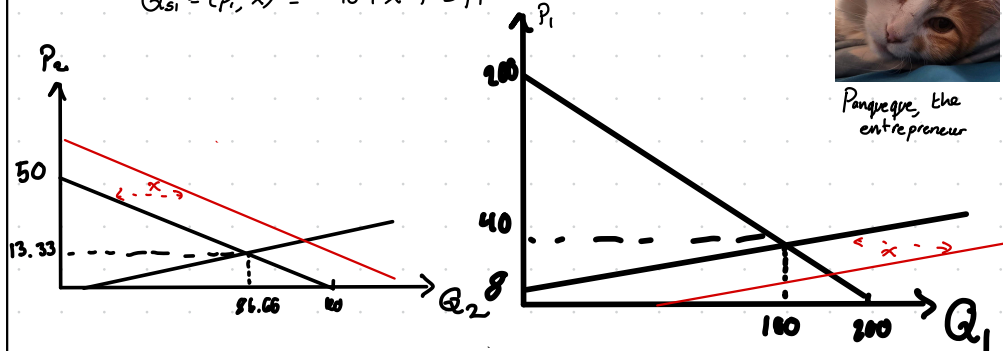
2.3 Pangque will buy honey from the second village and sell it in the first. We will solve a more general version.

• At any price P_2 , Pangque buys x units of honey $\rightarrow$ increase in demand

$$Q_{d2}(P_2, x) = 100 + x - 2P_2$$

• At any price P_1 , Pangque sells x units of honey $\rightarrow$

$$Q_{s1}(P_1, x) = -40 + x + 5P_1$$



Pangque, the entrepreneur

$$100 + x - 2P_2^* = 20 + 4P_2^*$$

$$\Rightarrow \frac{80+x}{6} = P_2^*(x)$$

$$200 - P_1^* = -40 + x + 5P_1$$

$$P_1^*(x) = \frac{240-x}{6}$$

If $x=30$ the new equilibrium prices are:

$$P_1^*(30) = 35 \quad ; \quad P_2^*(30) = \frac{55}{3} \approx 17.66$$

4) Therefore, Pangque gets $(P_1^*(x) - P_2^*(x))x = \left(35 - \frac{55}{3}\right)30 = \frac{50}{3} \cdot 30 = 500$

5) Pangque's Revenues are:

$$R(x) = (P_1^*(x) - P_2^*(x))x$$

$$= \left(\frac{240-x}{6} - \frac{80+x}{6}\right)x$$

$$= \left(\frac{240-x-80-x}{6}\right)x$$

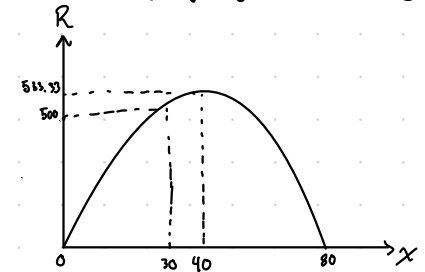
$$= \left(\frac{160-2x}{6}\right)x = \frac{80}{3}x - \frac{1}{3}x^2$$

Maximum revenues are obtained when:

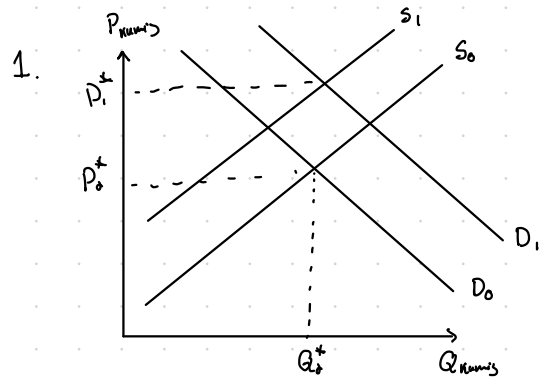
$$R'(x^*) = 0 \Rightarrow \frac{80}{3} - \frac{2}{3}x^* = 0 \Rightarrow 80 = 2x^*$$

$$\Rightarrow x^* = 40$$

$\therefore$ Pangque should buy and sell $x^* = 40$ units of honey, getting revenue of: $\frac{1600}{3} \approx 533.33$



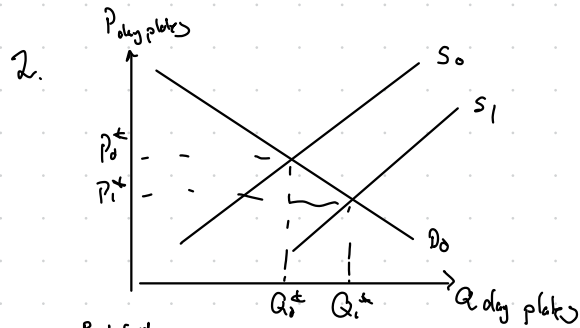
Problem 3



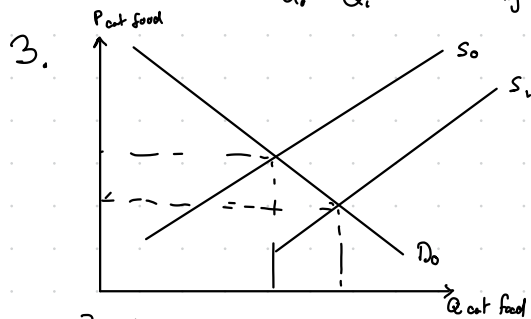
The equilibrium price rises.
The equilibrium quantity

- rises if D_1 moves more than S_1
- falls if S_1 moves more than D_1

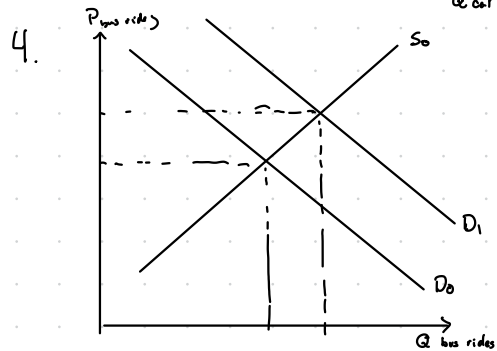
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$P^* \downarrow$, $Q^* \uparrow$



$P^* \downarrow$, $Q^* \uparrow$



$P^* \uparrow$, $Q^* \uparrow$